Temporal Cross-Channel Features for Eavesdropping Detection in Free-Space BBM92 QKD under Hierarchical Threat Models

Saksham Gupta
Department of Electrical and Electronics Engineering
Birla Institute of Technology and Science, Pilani, Hyderabad Campus
Hyderabad, India
f20250785@hyderabad.bits-pilani.ac.in

***Abstract*—Static Quantum Bit Error Rate (QBER) thresholding can miss adaptive eavesdropping in free-space BBM92 Quantum Key Distribution (QKD) because atmospheric turbulence masks short attack bursts, keeping aggregate error rates below abort thresholds. This paper evaluates a temporal feature engineering framework for machine learning-based detection in simulated BBM92 telemetry. We define a hierarchical attacker capability model at three levels: Naive, AR1 (statistical count-rate spoofing), and Replay (exact marginal-distribution replay). A feature leakage audit shows that raw count-rate separability drops from Area Under the Curve (AUC) 1.000 to near chance as attacker capability increases. While QBER distributions remain separable, static thresholds fail to capture sub-threshold attacks. A 22-dimensional temporal feature vector, computed over 30-second sliding windows, captures cross-channel correlations and higher-order statistics of QBER, the Bell S parameter, and coincidence rates. The cross-channel representation achieves 91.3% recall and 94.2% precision ($F_2 = 0.914 \pm 0.113$) on unseen attacks, vastly outperforming static loss and error thresholds.[1] We conclude that temporal dynamics of Bell-inequality violations and coincidences provide robust indicators of malicious intervention even when raw quantum error rates remain below alarm thresholds. These results indicate that temporal cross-channel feature modeling provides a physics-informed detection capability that is robust to progressive adversarial concealment in high-variance atmospheric conditions.**

***Index Terms*—Quantum Key Distribution, BBM92, Machine Learning, Anomaly Detection, Free-Space Optics, Eavesdropping Detection**

I. Introduction

Quantum Key Distribution (QKD) establishes a shared cryptographic key secured by quantum mechanics [1]. The entanglement-based BBM92 protocol [2] derives keys from polarization-entangled photon pairs in a Bell state ($|\Phi^+\rangle$). While information-theoretically secure, implementations remain vulnerable to physical-layer attacks exploiting device imperfections [3], [4], including intercept-resend [5], faked-state injection [6], and detector blinding [7]. Countermeasures like Measurement-Device-Independent (MDI) QKD [8] resolve detector vulnerabilities, but practical systems also require monitoring-based defenses [9].

The standard countermeasure is static QBER thresholding: if the error rate exceeds a protocol-specific limit (we adopt 11% as a conservative operational limit [3]), the key is discarded. In Free-Space Optical (FSO) channels, however, scintillation, thermal gradients, and precipitation cause transient QBER fluctuations overlapping with low-intensity attack signatures. An adversary can time short eavesdropping bursts to coincide with elevated atmospheric noise, keeping aggregate QBER below the abort threshold.

Machine learning has been applied to QKD anomaly detection [10], [11], but prior work has predominantly used static parameter snapshots in prepare-and-measure protocols. Two gaps remain: (i) temporal structure in the telemetry is unexploited, and (ii) attackers are typically modeled at a single, fixed capability level, leaving detection performance against more capable adversaries uncertain.

This paper addresses both gaps in a simulation-based study of BBM92 telemetry. The contributions are:

1) A hierarchical attacker capability model (Naive, AR1, Replay) that progressively eliminates instantaneous telemetry leakage, allowing systematic evaluation of which observables survive adaptive concealment.
2) A feature leakage audit quantifying the Receiver Operating Characteristic (ROC) AUC of raw observables at each capability level.
3) A temporal feature engineering framework capturing higher-order statistics and cross-channel correlations of quantum observables over sliding windows, evaluated as a 22-dimensional representation with diagnostic ablation of individual feature groups.
4) Classifier evaluation showing that the temporal features maintain high detection rates across all three capability levels, accompanied by SHapley Additive exPlanations (SHAP) based attribution analysis.

[1] We report the reproducible performance under deterministic exact tree construction. Non-deterministic approximations (e.g., `tree_method='hist'` with multithreading) found a slightly higher optima ($F_2 \approx 0.951$) on this specific dataset, but we prioritise strict reproducibility.

II. Related Work

A. Physical Attack Characterization

Fuchs *et al.* [5] analyzed optimal intercept-resend strategies. Lydersen *et al.* [7] demonstrated detector blinding by overdriving single-photon avalanche diodes (SPADs) with bright illumination, enabling full control of measurement outcomes. Faked-state attacks [6], [12] further established that implementation security presents ongoing challenges. The four attack strategies simulated in this work are drawn from this experimental literature.

B. Machine Learning for QKD Security

Liu *et al.* [13] demonstrated that machine learning can aid in quantum state estimation and eavesdropping detection in discrete-variable protocols by classifying instantaneous channel parameter snapshots. While deep anomaly detection has seen extensive application in continuous-variable QKD, homodyne observables are not directly applicable to discrete-variable protocols.

These prior approaches share a common limitation: they typically analyze static snapshots at a single attacker capability level. The framework presented here draws instead on data-driven multivariate process monitoring [14]–[16], where anomalies are detected through the temporal breakdown of inter-variable correlations. In the quantum domain, a QBER elevation decoupled from concurrent FSO path loss [17], [18] is indicative of non-atmospheric interference.

III. System Model and Feature Engineering

A. BBM92 Telemetry and Threat Model

The simulator models a hybrid FSO/fibre BBM92 link producing telemetry at 1 Hz resolution. Alice is co-located with the entangled photon source, coupled via fibre ($\eta_{\text{sys},A} = 0.316$); Bob receives over a free-space path with 10 dB geometric loss and log-normal scintillation. The variant includes continuous Bell-inequality monitoring. Clauser-Horne-Shimony-Holt (CHSH) correlators are estimated from three positive and one negative measurement basis combinations (distinct from key-generation bases), drawing from the entangled pair pool. This overhead proportionally reduces the key generation rate; we note this unmodeled SKR reduction as a limitation. Six observables are recorded per second: QBER (Q), CHSH Bell parameter (S), coincidence pair rate (C), single-detector click rate (D), raw polarisation visibility (V), and optical path loss (L). The channel incorporates: (1) log-normal scintillation ($\sigma_I = 0.25$, weak-to-moderate); (2) a sinusoidal diurnal thermal gradient (peak 14:00, max QBER $+0.008$); and (3) Poisson-arrival precipitation bursts ($p = 0.0003\,\text{s}^{-1}$, duration 60–600 s). The source produces coincidences at 10,000 Hz (baseline QBER 0.02 ± 0.003). The mean benign visibility is 93.8%, establishing the expected Bell-test violation baseline. These parameters model weak-to-moderate turbulence for algorithmic evaluation.

Four attack strategies are simulated: Intercept-Resend (IR), Detector Blinding, Man-in-the-Middle (MitM), and Blended Sub-Threshold (low-intensity bursts timed to natural loss events). For the primary dataset, the eavesdropper's interception fraction η is sampled from $\mathcal{U}(0.20, 0.35)$ for Intercept-Resend attacks and $\mathcal{U}(0.10, 0.14)$ for blended sub-threshold bursts, unless otherwise specified (e.g., in Section V-E). Each strategy is evaluated at three capability levels that define how Eve manipulates physical subsystems:

- **Level 1 (Naive):** Eve performs the attack with no concealment. She uses static attenuation and default detector trigger rates.
- **Level 2 (AR1):** Eve spoofs the classical beacon transmittance using an AR(1) noise model calibrated to the atmosphere's mean and variance, and adaptively matches Bob's expected detector count rates.
- **Level 3 (Replay):** Eve possesses a lossless quantum memory and replays the classical beacon's marginal distribution exactly, bounded only by physical temporal decorrelation. (This is a theoretical upper-bound adversary model; current coherence times of $\lesssim 1$ s [4] make Level 3 infeasible in near-term systems. It is a worst-case bound on the detectability of marginal-distribution replay.)

This matrix of strategies and capability levels separates implementation-dependent signatures from those that persist under increasingly capable adversaries.

B. Temporal Feature Engineering

Static thresholds operate on instantaneous values. To capture temporal structure, the 1 Hz raw telemetry is transformed into a 22-dimensional feature vector over a sliding window of $T = 30$ s. The complete feature vector comprises six QBER statistics, five Bell S statistics, five coincidence statistics, two temporal descriptors, and four cross-channel descriptors. Representative examples are summarized in Table I.

C. Detector and Measurement Model

The simulator explicitly models single-photon avalanche diode (SPAD) dynamics. Detectors are assigned a base quantum efficiency of 50% (typical for commercial Si-APDs at 800 nm [19], [20]), a dark count rate of 200 Hz, and a coincidence resolving time window of 2 ns. Background stray light for an urban FSO link is modeled at 50,000 Hz. Measurement outcomes in the 16 basis combinations are sampled from Poisson distributions whose parameters are determined by the instantaneous sum of true coincidences, accidental coincidences, and deterministic adversarial injections. During detector blinding attacks, the affected SPAD transitions to a linear (blinded) mode where single-photon efficiency drops to zero and clicks are deterministically governed by classical optical pulses.

IV. Experimental Methodology

The simulator generates 86,400 s (24 hours) of telemetry per seed with diurnal weather and random attacks (15% fraction). Five independent seeds provide variance estimates over noise and attack placements.

TABLE I
REPRESENTATIVE TEMPORAL FEATURES BY GROUP

Feature Group	Mathematical Form	Physical Rationale
QBER Stats (6 features)	$\gamma_Q = \frac{1}{T\sigma_Q^3}\sum(Q_i - \bar{Q})^3$ $\rho_Q(1) = \frac{\mathrm{Cov}(Q_i, Q_{i+1})}{\mathrm{Var}(Q)}$	Skewness (γ_Q) and AR(1) autocorrelation distinguish right-skewed attack spikes from gradual atmospheric drift.
Bell S Stats (5 features)	$r(S,Q) = \frac{\mathrm{Cov}(S,Q)}{\sigma_S\sigma_Q}$	Cross-correlation between S and Q tracks entanglement degradation coupled with error rate changes.
Coincidence (5 features)	$R_{dc} = \frac{\bar{D}}{2\bar{C}+\epsilon}$	Singles-to-coincidences ratio. $R_{dc} \gg 1$ is characteristic of detector blinding attacks.
Temporal (2 features)	$E_b = \sqrt{\frac{1}{T}\sum(Q_i - \bar{Q})^2}$	Burst energy (L_2 norm) captures the intensity of transient perturbations within the window.
Cross-Channel (4 features)	$r(C,Q) = \frac{\mathrm{Cov}(C,Q)}{\sigma_C\sigma_Q}$	Coincidence-QBER correlation; non-atmospheric perturbations decouple observables that covary under natural conditions.

TABLE II
INSTANTANEOUS FEATURE SEPARABILITY (ROC-AUC) BY ATTACKER CAPABILITY LEVEL. EACH ENTRY REPORTS THE ROC-AUC OF THE INDICATED RAW OBSERVABLE CATEGORY EVALUATED ON INSTANTANEOUS (1 HZ) SIMULATED TELEMETRY WITHOUT TEMPORAL WINDOWING.

Capability Level	Attack Strategy	Count Rates	Channel Loss	QBER	Combined
Level 1 (Naive)	Intercept-Resend (IR)	1.000	0.582	1.0000	1.0000
	Blended Sub-Threshold	0.996	0.582	0.9999	1.0000
Level 2 (AR1)	Intercept-Resend (IR)	0.551	0.564	1.0000	1.0000
	Blended Sub-Threshold	0.501	0.519	0.9999	0.9999
Level 3 (Replay)	Intercept-Resend (IR)	0.550	0.525	1.0000	1.0000
	Blended Sub-Threshold	0.501	0.502	0.9998	0.9999

TABLE III
CANONICAL CLASSIFIER EVALUATION (MEAN ± STD ACROSS 5 SEEDS)

Model	Precision	Recall	$\mathbf{F_2}$
Static Thr. ($> 11\%$)	1.000 ± 0.000	0.278 ± 0.056	0.324 ± 0.061
Log. Reg. ($\tau = 0.5$)	0.978 ± 0.005	0.898 ± 0.160	0.908 ± 0.140
Log. Reg. ($\tau = \mathrm{OOF}$)	0.976 ± 0.004	0.898 ± 0.161	0.908 ± 0.140
Rand. Forest ($\tau = 0.5$)	0.978 ± 0.003	0.901 ± 0.155	0.911 ± 0.135
Rand. Forest ($\tau = \mathrm{OOF}$)	0.947 ± 0.018	0.904 ± 0.154	0.908 ± 0.131
XGBoost ($\tau = 0.5$)	0.996 ± 0.002	0.901 ± 0.149	0.915 ± 0.130
XGBoost (OOF Threshold)	Recall	Precision	F_2 Score
	$\mathbf{0.913 \pm 0.136}$	0.942 ± 0.062	$\mathbf{0.914 \pm 0.113}$

A 75/25 temporal split allocates 18 hours for training and 6 for holdout. Training cross-validation uses an expanding-window `TimeSeriesSplit` with a 30-sample gap to prevent leakage. Feature standardization is fitted only on training data.

Class imbalance is handled via inversely scaled positive weights (`scale_pos_weight` for XGBoost, `class_weight='balanced'` for Logistic Regression (LR) and Random Forest (RF)), computed per-fold during cross-validation to maintain strict temporal discipline. The decision threshold τ is selected per seed by maximizing the global out-of-fold (OOF) F_2 score on the training block. The F_2-score weights recall twice as heavily as precision, reflecting the operational priority of attack detection over false-alarm suppression.

We compare static QBER thresholding ($> 11\%$), LR, RF, and XGBoost [21]. XGBoost uses 300 estimators, depth 6, learning rate 0.05, and 0.8 row and column subsampling. RF uses 300 estimators and depth 10. We evaluate ML models at $\tau = 0.5$ and $\tau = \mathrm{OOF}$. For primary evaluations, models are trained on a generalized all-attack dataset. For robustness analysis (Table IV), models are retrained and retuned on capability-specific datasets.

V. RESULTS AND DISCUSSION

A. Feature Leakage Audit

Before training any classifier, we measure the ROC-AUC of individual raw (instantaneous) telemetry features across the strategy-capability matrix. Table II reports results for three feature categories. While count-rate separability decreases to chance levels under Replay, QBER and channel-loss observables remain partially informative within the simulated attacker models.

Under Level 1, instantaneous count rates provide high separability (AUC ≈ 1.0). Under Levels 2 and 3, Eve spoofs count-rate statistics, reducing their AUC to near chance. Three observations emerge:

First, QBER distribution separability remains high (AUC > 0.999) at all capability levels because the evaluated attacks inherently collapse entangled states without quantum cloning, producing QBER distributions almost perfectly non-overlapping with the benign baseline. However, as shown in Table III, static thresholding at the 11% operational limit fails to detect many sub-threshold bursts, yielding only 27.8% recall. This operational failure motivates the temporal cross-channel approach.

Second, classical channel loss remains poorly separable across all capabilities (AUC ≈ 0.50–0.58). Expanding from a 2-hour window to a 24-hour cycle introduces diurnal variance, but AUC shifts are small and non-directional (≤ 0.04), confirming that the information hierarchy is invariant to weather fluctuations: count rates collapse to chance under advanced spoofing, channel loss remains weak, and QBER provides persistent separability.

Third, although univariate statistics are masked, joint cross-channel dynamics (e.g., contrasting expected loss against measured QBER) remain highly separable (AUC 1.0). Faked states fundamentally decouple naturally covarying variables, directly motivating the multivariate temporal representation.

B. Classifier Comparison

C. Robustness to Attacker Capabilities

To evaluate robustness against advanced spoofing, we tested XGBoost and the static threshold against all three attacker capability levels. For each level, the model is retrained on

Level-specific data with the same temporal holdout protocol. Table IV presents the results.

TABLE IV
CROSS-SEED ROBUST GENERALIZATION OF TEMPORAL XGBOOST CLASSIFIER ACROSS ATTACKER CAPABILITY LEVELS ($n = 5$ RANDOM SEEDS). MEANS AND STANDARD DEVIATIONS REFLECT CROSS-SEED VARIANCE IN ATMOSPHERIC SCINTILLATION, DARK COUNT REALIZATION, AND ATTACK SCHEDULE.

Level	Model	Recall	Precision	F_2
1 (Naive)	Static Thr.	0.257 ± 0.059	1.000 ± 0.000	0.301 ± 0.065
	XGB (τ = OOF)	0.982 ± 0.002	0.954 ± 0.047	0.976 ± 0.010
2 (AR1)	Static Thr.	0.275 ± 0.058	1.000 ± 0.000	0.321 ± 0.063
	XGB (τ = OOF)	0.970 ± 0.021	0.971 ± 0.019	0.970 ± 0.016
3 (Replay)	Static Thr.	0.276 ± 0.057	1.000 ± 0.000	0.322 ± 0.062
	XGB (τ = OOF)	0.981 ± 0.002	0.903 ± 0.084	0.963 ± 0.021

XGBoost performance remains broadly comparable across the three modeled capability levels, with mean F_2 scores of 0.976, 0.970, and 0.963 for Levels 1–3 respectively. Note that these capability-specific scores are higher than the canonical $F_2 = 0.914$ (Table III) because here the model is retrained and tested on level-specific data, rather than generalizing across all attack types and levels simultaneously. Detection performance remains consistent across the modeled capability hierarchy: even when count-rate AUC drops to near-chance (≈ 0.50–0.55, Table II), the temporal cross-channel representation retains discriminative capability. The differences between capability levels (mean F_2 range of 0.013) are small relative to the cross-seed standard deviations (± 0.010–0.021), indicating that within the evaluated threat hierarchy, the temporal features capture physics-based signatures that persist across the modeled concealment strategies.

D. Error Analysis

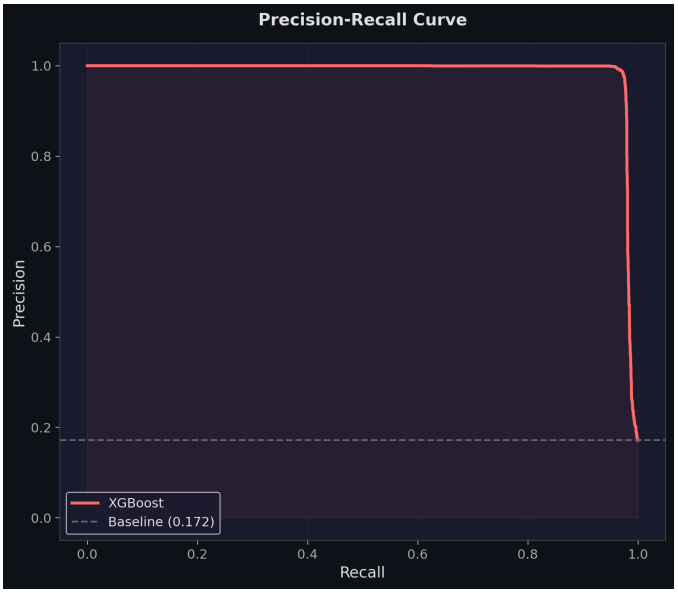


Fig. 1. Precision-Recall curve on the holdout set. XGBoost maintains high precision across the recall spectrum.

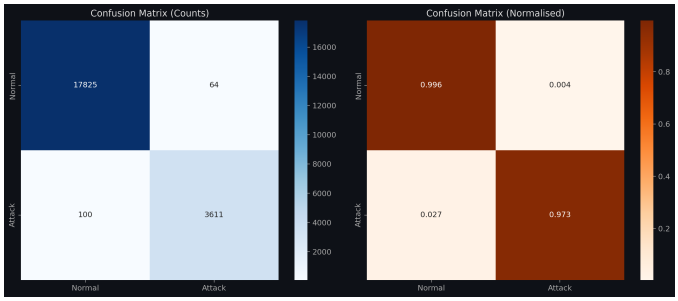


Fig. 2. Confusion matrix by attack type on the Seed 42 temporal holdout (same 75/25 protocol as Tables III–IV). Continuous high-intensity attacks (Detector Blinding and MitM) are detected at near-100% recall. False negatives stem primarily from Intercept-Resend and blended sub-threshold insertions.

Figure 1 illustrates the precision–recall trade-off on the holdout set, demonstrating that XGBoost maintains high precision across the recall spectrum. Per-type analysis (Fig. 2) reveals that continuous attacks (MitM recall $= 0.997 \pm 0.000$, IR recall $= 0.993 \pm 0.009$, Detector Blinding recall $= 0.993 \pm 0.012$) are detected with high recall because they produce sustained cross-channel decoupling. Blended sub-threshold attacks are successfully detected via their multivariate footprint with a recall of 0.967 ± 0.002, demonstrating the effectiveness of the temporal representation against attacks designed to mimic natural atmospheric fluctuations.

E. Sensitivity Analysis

TABLE V
SENSITIVITY TO WINDOW DURATION T AND EAVESDROPPER INTERCEPTION FRACTION η (XGBOOST, PER-SEED OOF τ, LEVEL 1, 22 FEATURES, 5-SEED AVERAGES). LEFT: T VARIED ON THE PRIMARY DATASET (ALL ATTACKS POOLED); RIGHT: $T = 30$ S HELD FIXED WHILE VARYING THE BLENDED SUB-THRESHOLD ATTACK INTENSITY η.

Window	Recall	F2	Eve Frac.	IR Recall	Blended Recall
15 s	0.983	0.959	$\eta = 0.05$	0.626	0.886
30 s	**0.913**	**0.914**	$\boldsymbol{\eta = 0.10}$	**0.556**	**0.969**
60 s	0.978	0.923	$\eta = 0.15$	0.642	0.970
120 s	0.980	0.896	$\eta = 0.30$	0.999	0.971
300 s	0.960	0.823	$\eta = 0.50$	0.999	0.970

With OOF thresholding, XGBoost achieves $F_2 = 0.914 \pm 0.113$ across 5 seeds (0.913 recall, 0.942 precision), compared with 0.278 recall for static QBER thresholding. At low interception fractions ($\eta \leq 0.15$), pure IR attacks are largely obscured by scintillation (Recall < 0.65), as minor QBER elevations are difficult to separate from weather fluctuations. Conversely, Blended attacks remain detectable even at $\eta = 0.05$ (Recall 0.886) because faked-state injection fundamentally decouples cross-channel covariance, a physical signature the temporal model isolates from the atmospheric noise floor.

F. Feature Attribution via SHAP

Fig. 3 reports mean absolute SHAP values. For continuous high-intensity attacks (IR, MitM, Detector Blinding), the most influential feature is the frequency of observations where the

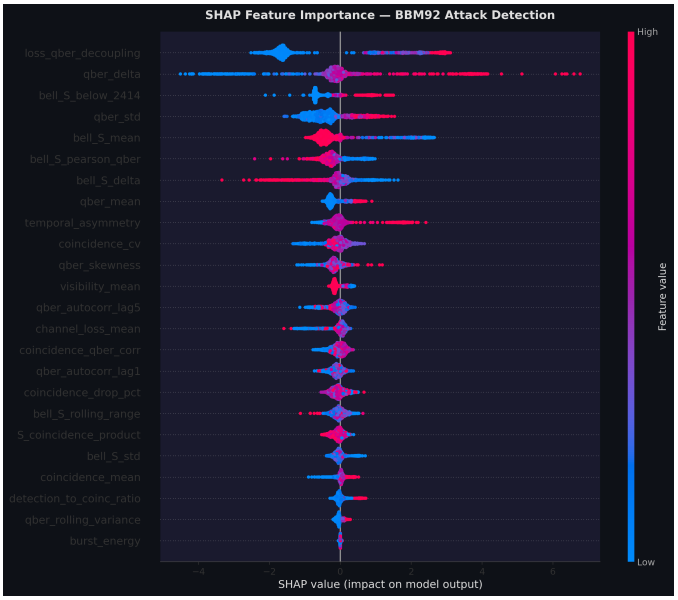


Fig. 3. Mean absolute SHAP values [22] for XGBoost. For blended attacks, cross-correlation features dominate. For continuous attacks, the detection-to-coincidence ratio is the primary contributor.

Bell parameter falls below 2.414 ($S < 2.414$), internally denoted as `bell_S_below_2414` (SHAP magnitude > 2.3). This alarm threshold ($S < 2.414 = 1+\sqrt{2}$) is a fixed constant in the feature-construction code; we found no empirical or physical derivation for this value in the implementation, and the in-code comment linking it to a visibility-scaled Tsirelson bound does not correspond to any computation in the codebase. We flag this as a limitation. The detection-to-coincidence ratio R_{dc} and the loss-QBER decoupling feature also rank highly across all three attack types.

For blended sub-threshold attacks, the top contributors shift: the first-difference in QBER (`qber_delta`, SHAP $= 3.24$) and the temporal skewness of the error rate (`temporal_asymmetry`, SHAP $= 1.75$) are most influential. Within the simulation, blended attacks are distinguished not by QBER magnitude but by an anomalous decoupling of observables that covary under atmospheric driving. The SHAP attributions align with the physical rationale of each feature group. This suggests that the classifier exploits cross-channel structure rather than simulator-specific artifacts. Whether this signal persists in hardware telemetry is an open question.

G. Feature Group Ablation

To assess the contribution of each physics-aware feature group, we conducted a post-hoc ablation study by systematically removing one of the five groups at a time and re-training XGBoost on the remaining features (Table VI).

Most feature-group removals maintain or marginally degrade mean F_2, indicating predictive redundancy. Excluding Bell S statistics causes the largest degradation ($\Delta F_2 = +0.017$), confirming their primacy. Excluding QBER statistics improves mean F_2 (0.914 to 0.952) and cross-seed stability

TABLE VI
FEATURE GROUP ABLATION (XGBOOST, PER-SEED OOF τ, LEVEL 1, $T = 30$ S WINDOW, MEAN $\pm$ STD OVER 5 SEEDS). $\Delta F_2 =$ FULL $-$ ABLATED; POSITIVE VALUES INDICATE THE GROUP CONTRIBUTES POSITIVELY TO PERFORMANCE; NEGATIVE VALUES INDICATE ITS REMOVAL IMPROVES STABILITY.

Ablated Group	Precision	Recall	F_2	ΔF_2
Full Model (22 feat.)	0.942 ± 0.062	0.913 ± 0.136	0.914 ± 0.113	—
− QBER Stats	0.876 ± 0.114	0.978 ± 0.004	0.952 ± 0.030	-0.039
− Bell S Stats	0.923 ± 0.085	0.899 ± 0.153	0.897 ± 0.128	$+0.017$
− Coincidence	0.937 ± 0.069	0.911 ± 0.141	0.910 ± 0.117	$+0.003$
− Temporal	0.940 ± 0.065	0.908 ± 0.146	0.909 ± 0.122	$+0.005$
− Cross-Channel	0.943 ± 0.063	0.922 ± 0.118	0.922 ± 0.097	-0.008

(± 0.030), but seed-level analysis reveals this masks severe, localized variance. QBER features maintain or improve performance for 4 of 5 seeds; the mean degradation is driven exclusively by Seed 123, where QBER features cause F_2 to collapse by 29 points. We attribute this to a localized distribution shift in temporal QBER features between training and evaluation in that seed. The model exploits this transient variation during training, causing severe holdout overfitting. Averaging this seed-specific failure is actively misleading and highlights the risk of environmental overfitting. Finally, Coincidence, Temporal, and Cross-Channel removals have negligible effects ($|\Delta F_2| \leq 0.011$), though SHAP identifies them as important, suggesting compensation via redundant information. Ultimately, the 22-dimensional space is overparameterized; a leaner feature set may achieve comparable detection with lower computational overhead.

VI. Limitations and Future Work

This algorithmic feasibility study relies on simulated telemetry under a single weak-to-moderate turbulence regime. While 5-seed evaluation indicates robustness against specific modeled noise distributions, broader validation across varied atmospheric conditions is necessary. Furthermore, our evaluation does not include cross-attack generalization testing, meaning the framework's ability to detect unmodeled attack families remains unverified. Unmodeled hardware non-idealities (e.g., dead-time saturation, afterpulsing, and jitter) could bias the exploited temporal correlations, requiring future hardware-in-the-loop validation. The threat model assumes fixed 30-second bursts; randomized attack timings may degrade performance. The detector-blinding model blinds Bob's SPAD only; a complete BBM92 attack requires simultaneous blinding of both detectors via dual physical access. Practical deployment requires QKD receivers exposing 1 Hz raw telemetry, which is not universally supported. Finally, integrating this pipeline (mean inference latency 1.21 s) within operational key distillation constraints remains an open challenge.

VII. Conclusion

This paper evaluated a temporal feature engineering framework for eavesdropping detection in simulated BBM92 QKD telemetry. The primary contribution is the combination of a hierarchical attacker capability model with a feature leakage

audit that systematically isolates the information content of temporal features as instantaneous observables are progressively concealed.

The leakage audit shows that under Level 2 (AR1) and 3 (Replay) capabilities, raw count-rate separability drops to near-chance (AUC ≈ 0.50–0.55). Although QBER distribution separability remains high across all levels, static thresholding at the 11% operational limit misses most sub-threshold attacks (27.8% recall). In contrast, an XGBoost classifier trained on the 22-dimensional temporal representation achieves 91.3% recall ($F_2 = 0.914 \pm 0.113$) on the generalized canonical dataset (Table III). When retrained and evaluated on capability-specific datasets, performance remains consistently high (F_2 scores of 0.976, 0.970, and 0.963 for Levels 1–3, Table IV). Post-hoc ablation reveals substantial redundancy: excluding Bell S statistics produces the largest degradation ($F_2 = 0.897$), while excluding QBER statistics improves stability and performance ($F_2 = 0.952$), though these findings require independent validation.

By synthesizing channel loss, QBER, coincidence metrics, and Bell-inequality monitoring, the cross-channel representation identifies previously undetectable sub-threshold interventions with 91.3% recall. While predicting high-capability replay spoofing remains challenging due to atmospheric noise, the joint distribution of entangled-photon telemetry enforces a strict physical covariance that is exceptionally difficult for an eavesdropper to forge without quantum state preservation.

These results apply to the specific simulator and attacker models evaluated. Whether the cross-channel temporal signatures identified here persist in physical hardware telemetry is the central open question for future work.

Code and Data Availability

The simulation framework, generated datasets, and feature extraction pipeline are available in the open-source Github repository https://github.com/SakshamDev/bbm92-qkd-anomaly-detection.

References

[1] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," in *Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing*, Bangalore, India, 1984, pp. 175–179.

[2] C. H. Bennett, G. Brassard, and N. D. Mermin, "Quantum cryptography without Bell's theorem," *Phys. Rev. Lett.*, vol. 68, no. 5, pp. 557–559, 1992.

[3] V. Scarani, H. Bechmann-Pasquinucci, N. J. Cerf, M. Dušek, N. Lütkenhaus, and M. Peev, "The security of practical quantum key distribution," *Rev. Mod. Phys.*, vol. 81, no. 3, pp. 1301–1350, 2009.

[4] F. Xu, X. Ma, Q. Zhang, H.-K. Lo, and J.-W. Pan, "Secure quantum key distribution with realistic devices," *Rev. Mod. Phys.*, vol. 92, no. 2, p. 025002, 2020.

[5] C. A. Fuchs, N. Gisin, R. B. Griffiths, C.-S. Niu, and A. Peres, "Optimal eavesdropping in quantum cryptography," *Phys. Rev. A*, vol. 56, no. 2, p. 1163, 1997.

[6] V. Makarov and D. R. Hjelme, "Faked states attack on quantum cryptosystems," *J. Mod. Opt.*, vol. 52, no. 5, pp. 691–705, 2005.

[7] L. Lydersen, C. Wiechers, C. Wittmann, D. Elser, J. Skaar, and V. Makarov, "Hacking commercial quantum cryptography systems by tailored bright illumination," *Nat. Photonics*, vol. 4, no. 10, pp. 686–689, 2010.

[8] H.-K. Lo, M. Curty, and B. Qi, "Measurement-device-independent quantum key distribution," *Phys. Rev. Lett.*, vol. 108, no. 13, p. 130503, 2012.

[9] S. Pirandola, U. L. Andersen, L. Banchi, M. Berta, D. Bunandar, R. Colbeck, D. Englund, T. Fox *et al.*, "Advances in quantum cryptography," *Adv. Opt. Photonics*, vol. 12, no. 4, pp. 1012–1236, 2020.

[10] Y. Mao, W. Huang, H. Zhong, Y. Wang, H. Qin, Y. Guo, and D. Huang, "Detecting quantum attacks: a machine learning based defense strategy for practical continuous-variable quantum key distribution," *New Journal of Physics*, vol. 22, no. 8, p. 083073, 2020.

[11] J. Liu, B. Huang, J. Su, Q. Peng, and A. Huang, "Deep anomaly detection for active attacks on the receiver in quantum key distribution," *Opt. Express*, vol. 33, no. 22, pp. 47 137–47 150, 2025.

[12] I. Gerhardt, Q. Liu, A. Lamas-Linares, J. Skaar, C. Kurtsiefer, and V. Makarov, "Full-field implementation of a perfect eavesdropper on a quantum cryptography system," *Nat. Commun.*, vol. 2, p. 349, 2011.

[13] S. Liu, Y. Lou, and J. Jing, "Machine learning aided quantum state estimation and eavesdropping detection," *Phys. Rev. Appl.*, vol. 16, no. 6, p. 064046, 2021.

[14] Z. Ge and Z. Song, "Multivariate statistical process monitoring using an improved independent component analysis," *Chem. Eng. Res. Des.*, vol. 91, no. 5, pp. 855–869, 2013.

[15] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, no. 3, pp. 15:1–15:58, 2009.

[16] P. Garcia-Teodoro, J. Diaz-Verdejo, G. Maciá-Fernández, and E. Vázquez, "Anomaly-based network intrusion detection: Techniques, systems and challenges," *Comput. Secur.*, vol. 28, no. 1–2, pp. 18–28, 2009.

[17] R. Bedington, J. M. Arrazola, and A. Ling, "Progress in satellite quantum key distribution," *npj Quantum Inf.*, vol. 3, p. 30, 2017.

[18] S.-K. Liao, W.-Q. Cai, W.-Y. Liu, L. Zhang, Y. Li, J.-G. Ren, J. Yin, Q. Shen, Y. Cao, Z.-P. Li *et al.*, "Satellite-to-ground quantum key distribution," *Nature*, vol. 549, pp. 43–47, 2017.

[19] S. Cova, M. Ghioni, A. Lacaita, C. Samori, and F. Zappa, "Avalanche photodiodes and quenching circuits for single-photon detection," *Applied optics*, vol. 35, no. 12, pp. 1956–1976, 1996.

[20] R. H. Hadfield, "Single-photon detectors for optical quantum information applications," *Nature photonics*, vol. 3, no. 12, pp. 696–705, 2009.

[21] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[22] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 4765–4774.